Identifying Key Indicators of Spam Emails: A GLM Approach

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Background

Which text characteristics influence whether an email will be classified as spam or not?

Objective: Optimize Spam Filters

Data Cleaning

```
# A tibble: 6 × 7
  crl.tot dollar  bang money  n000  make yesno
    <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
1     253  0.189 0.882  0     0.37  0    y
2     112  0.08  0      0     0     0    n
3      17  0     0.5    0     0     0    y
4     319  0     0.141  0     0     0    n
5     513  0.464 0.406  0     0.66  0.33 y
6    2049  0.157 0.502  0.08  0.42  0    y
```

Response (Y): yes/no (Spam vs. Non-spam)
Predictors (X): 6 text-based features

Data summary

Name	spam_clean
Number of rows	920
Number of columns	7
_____	
Column type frequency:	
factor	1
numeric	6
_____	
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
yesno	0	1	FALSE	2	n: 557, y: 363

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
crl.tot	0	1	265.76	502.61	1	35.75	89.5	240.50	3498.00	▇▁▁▁
dollar	0	1	0.07	0.26	0	0.00	0.0	0.03	5.30	▇▁▁▁
bang	0	1	0.24	0.55	0	0.00	0.0	0.30	7.84	▇▁▁▁
money	0	1	0.07	0.30	0	0.00	0.0	0.00	6.66	▇▁▁▁
n000	0	1	0.09	0.33	0	0.00	0.0	0.00	4.32	▇▁▁▁
make	0	1	0.12	0.36	0	0.00	0.0	0.00	4.34	▇▁▁▁

Exploratory Data Analysis (EDA)

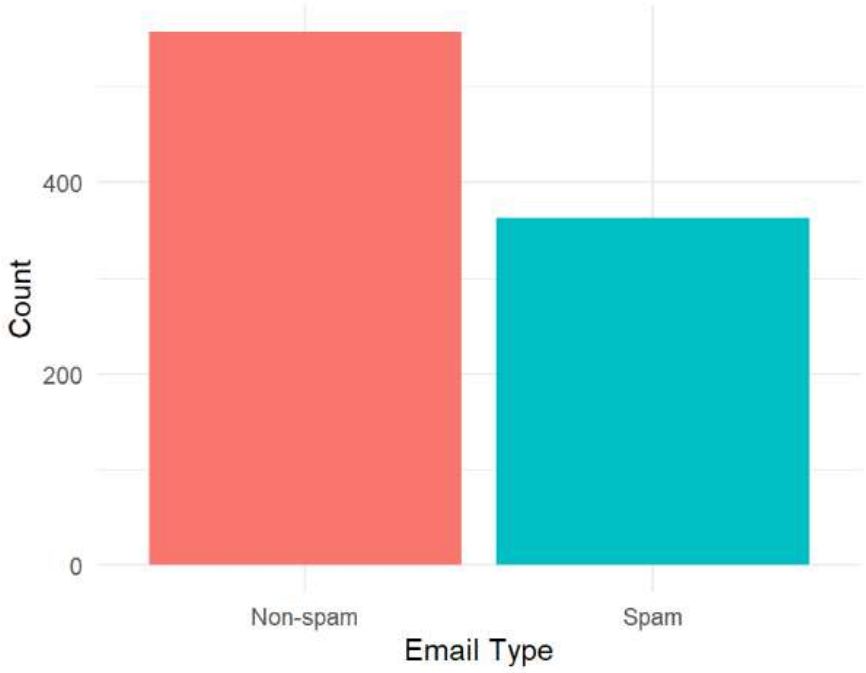


Table 1: Grouped summary of predictors by email type.

yesno	variable	mean	median	zero_pct	max
n	bang	0.074	0.000	75.9%	3.571
n	crl.tot	168.259	55.000	0.0%	3467.000
n	dollar	0.010	0.000	89.6%	0.427
n	make	0.101	0.000	84.2%	4.340
n	money	0.005	0.000	98.7%	0.800
n	n000	0.006	0.000	97.1%	0.960
y	bang	0.490	0.298	20.1%	7.843
y	crl.tot	415.372	171.000	0.0%	3498.000
y	dollar	0.158	0.051	43.8%	5.300
y	make	0.145	0.000	67.8%	2.350
y	money	0.177	0.000	66.4%	6.660
y	n000	0.219	0.000	71.1%	4.320

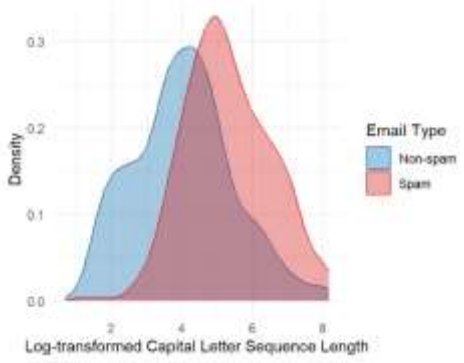
Sample Size:

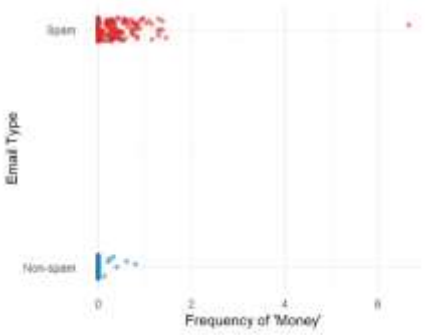
Total Observations: n = 920

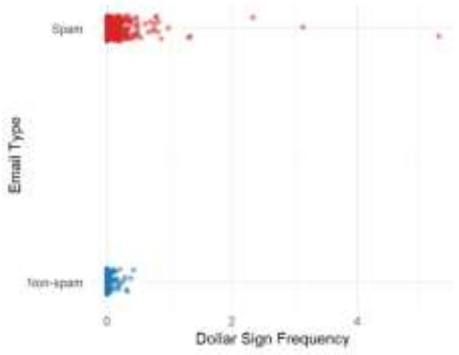
Non-Spam: n_1 = 557 (60.5%)

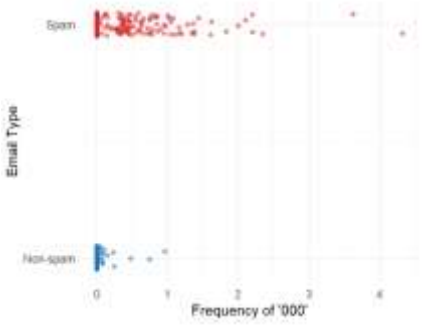
Spam: n_2 = 363(39.5%)

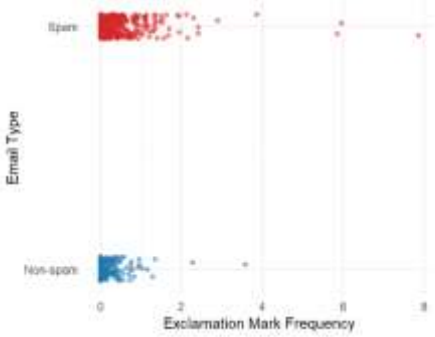
Exploratory Data Analysis (EDA) - Key Findings

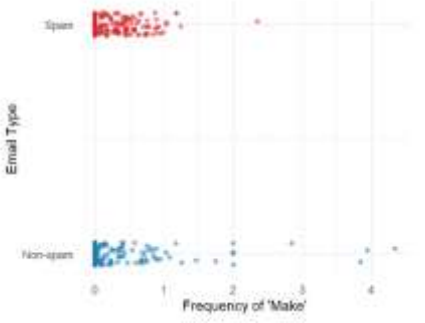


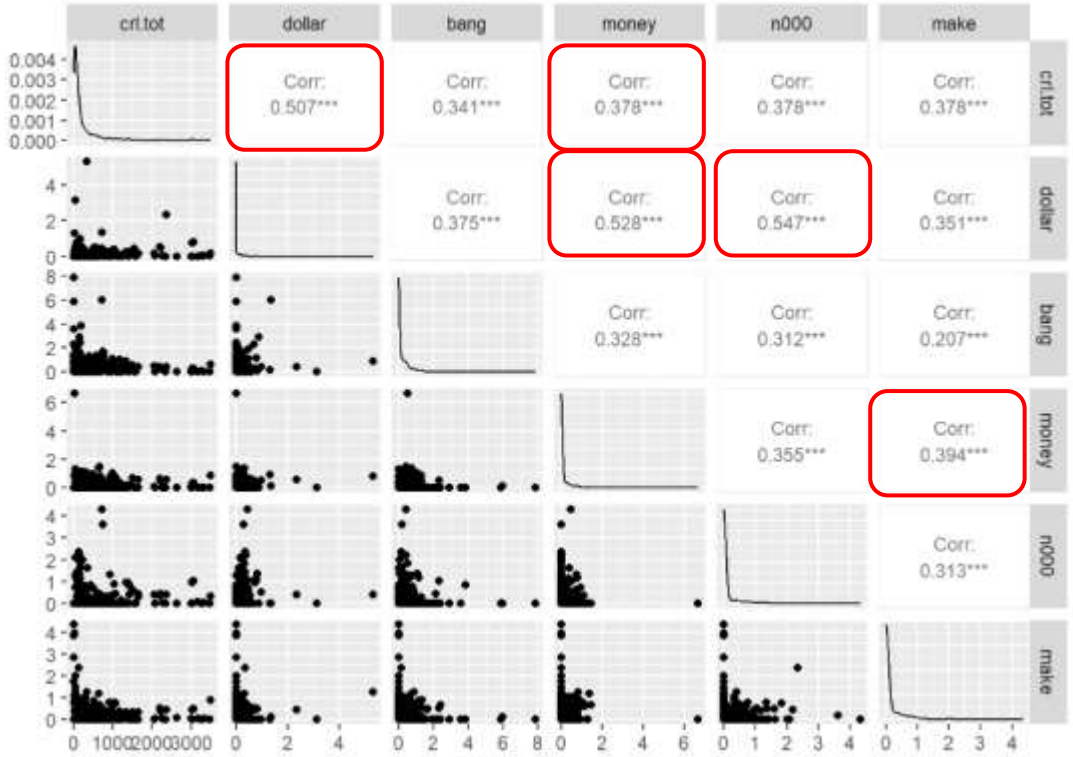












Spam Indicators: Strong patterns observed in **dollar, bang,** and **money**.

Key Feature: String **n000** shows a distinct frequency spike in spam.

Weak Pattern: Variable **make** appears less discriminative.

Model Suitability: Low multicollinearity; variables ready for Logistic Regression.

Statistical Analysis: Logistic Regression

$$\ln\left(\frac{\mathrm{p}}{1-\mathrm{p}}\right) = \beta_0 + \beta_1\ (\mathrm{crl.tot}) + \beta_2(\mathrm{dollar}) + \beta_3(\mathrm{bang}) + \beta_4(\mathrm{money}) + \beta_5(\mathrm{n000}) + \beta_6(\mathrm{make})$$

- p is the probability that the email is spam.
- $\ln\left(\frac{\mathrm{p}}{1-\mathrm{p}}\right)$ is known as the log-odds.
- β_0 is the intercept, and $\beta_1 \ldots \beta_6$ are the regression coefficients for each variable.

Model Fitting

```
Call:
glm(formula = yesno ~ crl.tot + dollar + bang + money + n000 +
    make, family = binomial(link = "logit"), data = spam_clean)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.8101190  0.1254636 -14.427  < 2e-16 ***
crl.tot      0.0005502  0.0001886   2.917 0.003533 **
dollar       8.1346140  1.5396484   5.283 1.27e-07 ***
bang         2.9172085  0.3363971   8.672  < 2e-16 ***
money        5.9724851  1.2455257   4.795 1.63e-06 ***
n000         3.4827736  1.0261134   3.394 0.000688 ***
make        -0.4553154  0.4065463  -1.120 0.262731
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1234.17  on 919  degrees of freedom
Residual deviance:  752.44  on 913  degrees of freedom
AIC: 766.44

Number of Fisher Scoring iterations: 7
```

Model Selection & Results

Full : yesno ~ crl.tot + dollar + bang + money + n000 + make

Reduced : yesno ~ crl.tot + dollar + bang + money + n000

Selection: Reduced model via backward elimination.

- Full Model AIC: 766.44
- Reduced Model AIC: 766.04

Improved parsimony without sacrificing predictive power.

	OR	2.5 %	97.5 %
(Intercept)	0.158	0.123	0.200
crl.tot	1.001	1.000	1.001
dollar	3573.436	202.301	85411.590
bang	18.819	10.013	37.592
money	341.550	38.501	5244.916
n000	30.794	5.218	301.686

	Actual	
Predicted	n	y
n	523	122
y	34	241

- Overall Accuracy: 83.04 %
- False Positive Rate: 6.1 %
- Sensitivity: 67.4%

dollar, **money** are highly significant.
n000, **bang** are significant.
Capital letters have a negligible effect on predictions, or possibly no effect at all.

Model Validation & Performance

80/20 Train-Test

```
Call:
glm(formula = yesno ~ crl.tot + dollar + bang + money + n000,
    family = binomial(link = "logit"), data = spam_train)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.0268297  0.1439189 -14.083  < 2e-16 ***
crl.tot      0.0005845  0.0001947   3.002  0.00268 **
dollar       8.6024844  1.8841848   4.566 4.98e-06 ***
bang         3.1467016  0.3877579   8.115 4.85e-16 ***
money        8.6141664  1.9143086   4.500 6.80e-06 ***
n000         2.7695467  1.0437369   2.653  0.00797 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 968.57  on 735  degrees of freedom
Residual deviance: 565.88  on 730  degrees of freedom
AIC: 577.88

Number of Fisher Scoring iterations: 8
```

	Actual	
Predicted	n	y
n	85	35
y	7	57

Final Test Accuracy: 77.17 %
False Positive Rate (Unseen Data): 7.61 %

Conclusion, Future Work & Limitations

Primary effect variables:

dollar ($P < 0.001$, OR ≈ 3573), **money** ($P < 0.001$, OR ≈ 341)

Secondary effect variables:

n000 ($P < 0.01$, OR ≈ 30.8), **bang** ($P < 0.001$, OR ≈ 18.8)

Weak effect Variables:

crl.tot ($P < 0.01$ (**), OR ≈ 1.001)

Removing non-significant variables (e.g., make) achieved a more parsimonious and efficient filter.

Performance:

83% Accuracy confirms text-based features are effective for real-world baseline filtering.

Conclusion, Future Work & Limitations

Limitations:

Model serves as a **strong baseline**, not a complete solution.

Future Enhancements:

1. Integrate advanced **Natural Language Processing (NLP).**
2. Test non-linear methods (e.g., **Random Forest**).
3. Expand feature set beyond simple frequencies.

Q & A

Thanks for your listening!